Caleb Traxler

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Professional Summary

Data Science graduate student specializing in Machine Learning, Data Science and Computer vision. Published academic researcher with significant entrepreneurial experience developing scalable, Artificial Intelligence systems. Actively seeking PhD opportunities in computer science – start date of September 2026. Actively seeking job opportunities in the data science and machine learning space – start date of December 2025.

Education

- **UC Irvine, Master of Science in Data Science (GPA: 3.97)** Sep 2024 – Dec 2025
  Specializations: AI, Generative Models, Computer Vision, GIS, Big Data *Irvine, CA*
- **UCLA, Bachelor of Science in Mathematics & Computer Science (GPA: 3.81)** Sep 2022 – Jun 2024
  Honors: Phi Theta Kappa, AI Safety Fellowship *Los Angeles, CA*

Research Experience

- **Graduate Research Assistant, UC Irvine (Prof. Erik Sudderth)** Jun 2025 – Present
  Variational inference with Gauss-Markov distributions. Applying this to SDE time series models, as well as a potential future project involving chemical data. *Irvine, CA*
- **Graduate Research Assistant, UC Irvine (Prof. Jun Wu)** Jun 2025 – Present
  Satellite data from ECOSTRESS, examining extreme heat distributions across urban zones in California cities. Emphasis placed on spatial data handling, GIS, and Python-based geospatial analysis. *Irvine, CA*
- **Undergraduate Researcher, UCLA Mathematics (Prof. Shbia Biswal)** Mar 2023 – Jun 2023
  Modeled COVID-19 dynamics using extended SIR/SEIR models, published in arxive *Los Angeles, CA*

Industry Experience

- **Founder & CEO, Traxler Technology LLC** Nov 2024 – Present
  Founded and lead an AI-focused startup building multimodal intelligence systems. Designed and deployed scalable full-stack platforms using React, Firebase, AWS EC2, Vercel, and Auth0.(Website) *LA, CA*
- **Master of Data Science Program Ambassador, UC Irvine** Sep 2024 – Present
  Ambassador for the Masters of Data Science program at UC Irvine 2024-2025 cohort; I support the creation and development of ideas for marketing campaigns. (UCI Instructors and Staff) *Irvine, CA*
- **Data Scientist & ML Engineer Intern, Amgen** Jul 2024 – Sep 2024
  Data analysis and generative AI models for rare disease prioritization. Turned year long process into minute long pipeline using genAI. Build streamlit application to visualize data for stakeholders. Winner: Amgen AI Symposium 2024 *Remote*

Technical Projects

- **AI Life Journal using VLMs and LangChain Memory** June 2025 – Present
  Developing a cross-platform mobile application (Android & iOS) that allows users to capture, organize, and reflect on their daily experiences using Meta Ray-Ban AI glasses and other life-logging devices. Integrates advanced vision-language models (VLMs) and LangChain to deliver memory-augmented AI summaries and intelligent journaling features. Responsible for end-to-end architecture, including media ingestion, secure storage, and AI-driven insights. (GitHub)
- **Generative Route Prediction with HMMs and 3D Point Clouds** May 2025 – June 2025
  Developed discrete Hidden Markov Models (HMMs) to generate realistic driving routes from KITTI-360 3D point cloud maps. Currently under review for future publication.(GitHub)
- **Multivariate Statistical Analysis of Exoplanet Habitability** May 2025 – June 2025
  Comprehensive multivariate statistical analysis of 517 exoplanets from the NASA Exoplanet Archive to identify potentially habitable worlds and quantify detection bias in current surveys. Interactive 3D visualization using Streamlit and PyDeck to analyze habitability trends (Live App) . Published in arxive.

- **Autonomous Driving via CNN and Groq API** Sep 2024 – March 2025
  Real-time lane detection and multimodal scene analysis using U-Net CNN; Achieved >95% accuracy and <50ms latency. Currently under review for future publication.(GitHub)
- **Real Estate ROI Geo-Locator** Jan 2025 – March 2025
  Interactive 3D visualization using Streamlit and PyDeck to analyze real-time ROI trends (Live App)

Publications

- **Traxler, C.**, et al. (2025). Multivariate Statistical Analysis of Exoplanet Habitability: Detection Bias and Earth Analog Identification. *arXiv:2506.18200*
- **Traxler, C.**, et al. (2025). Analysis of COVID-19 Infection Dynamics: Extended SIR Model Approach. *arXiv:2505.13754*

Technical Skills

- **Languages & Cloud:** Python, R, JavaScript (React), SQL, MATLAB, Bash, AWS (EC2, S3), Firebase, Vercel, Docker, Streamlit, Git/GitHub
- **Frameworks & Libraries:** NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Keras, OpenCV, Seaborn, Matplotlib, Hugging Face, LangChain, Plotly, PyDeck
- **Machine Learning & AI:** Deep Learning, Generative AI, Vision-Language Models (VLMs), Transformers, NLP, Reinforcement Learning, HMMs, Variational Inference, Bayesian Networks
- **Big Data & Scientific Computing:** Hadoop, Spark, GIS, ECOSTRESS, rasterio, shapely, SciPy, Statsmodels
- **Dev Tools & Infra:** Jupyter, VSCode, REST APIs, Linux CLI, CI/CD (GitHub Actions), Auth0

Interests and Hobbies

- Active investor in real estate (homes in the mid-west) and public markets; passionate about financial modeling and long-term portfolio strategy
- Deeply interested in augmented reality, human cognition, and the intersection of AI and wearable computing
- Enjoys programming full-stack applications, exploring new machine learning architectures, and contributing to open-source projects
- Committed to academic research and paper publication in AI, statistical modeling, and multimodal systems
- Personal interests include surfing and weight training.